

DIONIS LUDOVICK NASUWA

Software Engineer & Data Analyst | Distributed Systems & Data Infrastructure

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PROFESSIONAL SUMMARY

Computer Science graduate from the University of Dar es Salaam (UDSM) with a strong foundation in systems architecture. Focused on combining software engineering and data analytics to build enterprise-scale solutions and evaluate modern data stacks. Certified in Full Stack Open (Grade 5/5) and Data Analytics, having built distributed backend platforms and local LLM RAG pipelines. Dedicated to academic and technical growth through graduate-level research in Data Science and Software Engineering.

EDUCATION & CERTIFICATIONS

BSc in Computer Science — University of Dar es Salaam (CoICT)	Expected Oct 2026
Full Stack Open Certification — University of Helsinki (Grade 5/5 Maximum Score)	July 2026
Data Analytics Essentials — Cisco Networking Academy	Sept 2026
GDPR Foundations Certification	Sept 2026

Key Coursework: Data Structures & Algorithms, Distributed Systems, Database Systems (SQL), Operating Systems & Linux Admin, Computer Networks, Systems Programming.

TECHNICAL EXPERIENCE

Data Analyst (HQ) • Halotel (Viettel Tanzania Plc) Sept 2026 – Present | Dar es Salaam

- Evaluating and prototyping modern enterprise data stacks, leveraging **SQL**, **PostgreSQL**, and **Python** to analyze large-scale telecommunications data infrastructure.
- Researching advanced analytics and visualization workflows utilizing **Tableau**, **DuckDB**, and **Apache Superset** to modernize internal data reporting pipelines.

Software Engineering Collaborator • Mainstream Group (Technology Division) March 2026 – May 2026 | Dar es Salaam

- Engineered modular, service-oriented backend features using **TypeScript**, **Node.js**, and **Express/Fastify**, speeding up API performance and ensuring seamless data flow between services.
- Built secure **RESTful API endpoints** and optimized SQL data access layers, actively participating in bi-weekly Agile/Scrum sprint cycles, rigorous PR code reviews, and production release pipelines.

Software Development Trainee • UDICTI / StartHub Africa Partnership July 2025 – Sept 2025 | Dar es Salaam

- Developed the **Smart Cargo** cross-platform logistics platform using **Flutter**, **Django**, and **PostgreSQL**, connecting mobile clients to automated dispatch and cargo telemetry services.
- Implemented dynamic multi-factor pricing algorithms (distance matrix, weight tiering, fragility index) integrating **OpenStreetMap** routing APIs with bilingual Swahili/English localization.

IT Infrastructure Trainee • TPC Limited July 2024 – Sept 2024 | Kilimanjaro

- Configured managed network switches, deployed routers, and set static IP bindings to optimize enterprise LAN uptime and resolve hardware/OS diagnostics across enterprise workstations.

SELECTED TECHNICAL PROJECTS

Plaqode Platform — Distributed Monorepo Architecture for Digital Identity & QR Solutions

TypeScript • Next.js 14/15 • Fastify • Turborepo • Prisma ORM • Neon PostgreSQL • RS256 JWT • Docker

- Architected an 8-application monorepo using **Turborepo** and **pnpm workspaces**, decoupling digital-card, QR analytics, and ticketing domains; enabled up to 20 concurrent tasks while reducing redundant builds through remote caching.
- Designed a stateless **RS256 JWT** authentication service with **Fastify**, centralizing token issuance while enabling frontend applications to independently verify public-key signatures; eliminated cross-service authentication calls while preserving HttpOnly cookie isolation across **.plaqode.com** subdomains.
- Implemented isolated persistence boundaries using **Neon PostgreSQL** and dedicated **Prisma** clients per microservice, reducing schema coupling and enabling independent database migrations across service domains.
- Built a high-throughput QR analytics engine using **Fastify**, **Sharp** image processing, and **GeoIP tracking** over denormalized PostgreSQL tables; optimized telemetry aggregation to achieve sub-100 ms scan-query response times.

INTLEQ — Local LLM RAG & Vector Search System

Python • FastAPI • Ollama • pgvector • PostgreSQL • React/Vite • Docker

- Built an offline-first RAG platform using **Ollama (llama3.2:1b)** and **pgvector** embeddings, enabling private document Q&A without cloud API dependencies or external data egress.
- Engineered an asynchronous **FastAPI** backend using **asyncpg** and **pgvector** cosine-similarity search, achieving sub-second vector retrieval across concurrent user sessions.

Strata — PHP Legacy Modernization & Code Analysis Engine

Python 3.11+ • AST Parsing • Graph Theory (SCC) • Streamlit UI • Neo4j • Docker

- Engineered a static-analysis engine that transforms legacy PHP monoliths into **Abstract Syntax Trees (ASTs)** and dependency graphs, enabling architectural risk assessment for refactoring and microservice migration.
- Implemented **Strongly Connected Components (SCC)** clustering and interactive **Streamlit** visualizations to identify circular dependencies and calculate graph-centrality risk scores, enabling analysis of tightly coupled legacy components.

TECHNICAL SKILLS & COMPETENCIES

Languages: TypeScript, JavaScript (ES6+), Python, SQL, Dart, Bash

Frontend & UI: React, Next.js (App Router), Tailwind CSS, Redux Toolkit, Vite, Streamlit, HTML5/CSS3

Backend & APIs: Node.js, Fastify, Express.js, FastAPI, Django, RESTful APIs, Asymmetric RS256 JWT, Microservices

Data & Infrastructure: PostgreSQL, DuckDB, pgvector, Apache Superset, Tableau, Prisma ORM, Redis

DevOps & Tools: Docker, Turborepo, Git, GitHub Actions, Linux (Ubuntu/Debian), Nginx, Postman

Methodologies & CS: Object-Oriented Analysis (OOAD), AST & Graph Theory, Data Governance & Privacy (GDPR), Data Modeling